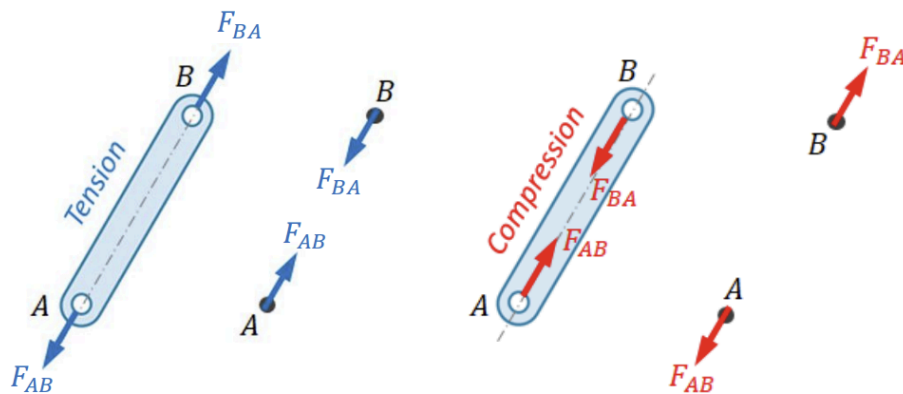


Static Analysis of Trusses

Trusses are structures meant to support static loads, consisting of straight linkages with pin joint connections between each other. This means that the forces experienced by each member is along its length, as each member is a two force body, hence each member experiences either tension, compression, or no load.

External loadings are only applied as point loads on the pin joints only, and other forces or distributed loads can be expressed in terms of point loads on the pin joints.

For a member, the axial forces acting on the member are equal and opposite to the forces acting on the pin, being an action-reaction pair.



To analyse trusses, there are two methods, method of joints and method of sections.

Method of Joints

Firstly, find any external reactions by treating the whole truss as a rigid body.

Then, consider the axial loads experienced by each member by drawing an FBD of each member and pin

This gives FBDs for each pin, which allow us to analyse each pin with static equilibrium methods to solve for each force in each member. Typically, each member is assumed to be in tension, hence the reaction force acting on the pin will be away from the pin itself.

Hence, by starting at a joint with only 2 unknowns, and working on to the next joints, the forces in the member can be found.

Zero Force Members

Method of joints is simple, but tedious. To speed up this method, we can remove unnecessary members that do not undergo load.

There are three cases in which a member has no load:

1. Two members connected at one pin joint with no external load
 - a. both members are zero force members
2. Two members connected at one pin joint with an external load aligned to one member
 - a. member not aligned to load is zero force member
3. Three members connected at one pin joint with no external load, where two members are aligned.
 - a. member not aligned to the other two is a zero force member.

As long as one joint of a member satisfies either one of these conditions, it is a zero force member.

Zero force members are also only specific to one loading condition.

Method of Sections

Unlike method of joints which focuses on concurrent loads, method of sections considers a section of the truss structure to apply non-concurrent equilibrium equations

If we need only some of the member's forces, we can apply the method of sections, by considering only a section of the truss structure, and accounting for the forces by other trusses and the external loads, while not concerning the internal loads in the truss structure.

Sometimes, method of joints may be needed after finding one force from method of sections.